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ARTICLE



Effect of different velocity loss thresholds during a power-oriented resistance training program on the mechanical capacities of lower-body muscles

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ABSTRACT

This study compared the effects of two velocity loss thresholds during a power-oriented resistance training program on the mechanical capacities of lower-body muscles. Twenty men were counter-balanced in two groups (VL10 and VL20) based on their maximum power capacity. Both groups used the same exercises, relative intensity and repetition volume, only differing in the velocity loss threshold of each set (VL10: 10% vs. VL20: 20%). Pre- and post-training assessments included an incremental loading test and a 15-m linear sprint to assess the force- and load-velocity relationships and athletic performance variables, respectively. No significant between-group differences ($P > 0.05$) were observed for the force-velocity relationship parameters (ES range = 0.15–0.42), the MPV attained against different external loads (ES range = 0.02–0.18) or the 15-m sprint time (ES = 0.09). A high between-participants variability was reported for the number of repetitions completed in each training set (CV = 30.3% for VL10 and 29.4% for VL20). These results suggest that both velocity loss thresholds induce similar changes on the lower-body function. The high and variable number of repetitions completed may compromise the velocity-based approach for prescribing and monitoring the repetition volume during a power-oriented resistance training program conducted with the countermovement jump exercise.

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KEYWORDS

Force-velocity relationship;
load-velocity relationship;
velocity-based training;
training volume;
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Introduction

A key physical performance factor in numerous sport activities is the ability to produce a high amount of force in the shortest time possible (Haff & Nimphius, 2012). Success in such actions (e.g., jumping, sprinting, change of direction, etc.) has been closely related to high muscle power outputs, that is, to the ability to produce high levels of force at high muscle contraction velocities (Cronin & Hansen, 2005; Swinton, Lloyd, Keogh, Agouris, & Stewart, 2014). Ballistic exercises, which are characterized by accelerating the body mass or an implement through the whole range of motion, are commonly included in resistance training (RT) programs aiming to enhance maximal power production (Cormie, McGuigan, & Newton, 2011; Haff & Nimphius, 2012). Therefore, of particular interest for coaches, strength and conditioning professionals and sport scientists would be to identify the most appropriate approaches for prescribing and monitoring RT programs aiming to maximize power performance that typically include ballistic exercises (e.g., vertical jumps).

Velocity-based training (VBT) has been proposed as an objective and practical method of prescribing both the intensity and volume of RT programs (González-Badillo & Sánchez-Medina, 2010), which are considered the two acute RT variables with most influence in the training response (Kraemer & Ratamess, 2004; Scott, Duthie, Thornton, & Dascombe, 2016). Exercise intensity has been traditionally prescribed as a

percentage of the one-repetition maximum (%1RM) (González-Badillo & Sánchez-Medina, 2010; Kraemer & Ratamess, 2004). However, the strong and stable relationship observed between movement velocity and %1RM supports the use of movement velocity to prescribe the relative load (Conceição, Fernandes, Lewis, González-Badillo, & Jimenez-Reyes, 2016; González-Badillo & Sánchez-Medina, 2010; Sánchez-Moreno, Rodríguez-Rosell, Pareja-Blanco, Mora-Custodio, & González-Badillo, 2017). More recently, the velocity loss incurred during a training set has been proposed as a criteria to decide when each set should be stopped (Pareja-Blanco et al., 2016; Pareja-Blanco, Sánchez-Medina, Suárez-Arrones, & González-Badillo, 2016). This approach is justified by the significant associations observed between the magnitude of velocity loss and the degree of fatigue (Sánchez-Medina & González-Badillo, 2011). Accordingly, it would be expected that the degree of velocity loss experienced during the training sets influence the RT effect. However, there is scarce evidence regarding the effect of different velocity loss thresholds (e.g., 10% vs. 20%) on the neuromuscular adaptations following a power-oriented RT program.

To date, only two studies have analysed the effect of different velocity loss thresholds during RT on the neuromuscular adaptations (Pareja-Blanco et al., 2016, 2016). Both studies used the full-squat exercise and high intensity oriented loads (i.e. 50–85%1RM). While similar increases in maximal strength were observed for the different velocity loss

thresholds, ballistic performance (i.e. unloaded countermovement jump [CMJ] height) further improved in the groups that used lower velocity loss thresholds (i.e., minimizing fatigue). It should be noted that in both studies the groups that used a high-velocity loss threshold (30% and 40%) performed a significantly higher training volume (i.e., higher number of repetitions) than the groups that used a low velocity loss threshold (15% and 20%). Therefore, since RT muscle responses are markedly influenced by the training volume (González-Badillo, Izquierdo, & Gorostiaga, 2006; Schoenfeld, Ogborn, & Krieger, 2017), it could be interesting to explore the effect of different velocity loss thresholds when the total training volume is equalised between groups. It is also unknown whether VBT can be a suitable approach for prescribing and monitoring a power-oriented RT program with lower relative intensities ($< 50\%1RM$) and using ballistic exercises (e.g., the loaded CMJ), which are commonly recommended to develop maximum power and high-velocity explosive actions performance. In this regard, since ballistic exercises are commonly performed with minimal fatigue (Cormie et al., 2011), it seems reasonable to explore in these exercises the effects of lower velocity loss thresholds than the typically used in previous studies.

To address the existing gaps in the literature, the objective of the present study was to compare the effect of two velocity loss thresholds (10% vs. 20%) during a power-oriented RT program with the same exercise intensity and repetition volume on the mechanical capacities of lower-body muscles. We hypothesized that the RT program allowing a 10% velocity loss would result in greater short-term power performance improvements than a 20% velocity loss due to the greater number of repetitions performed at higher velocity. The expected findings should contribute to refining the VBT approach to prescribe intensity and volume of power-oriented RT programs.

Method

Participants

Twenty men volunteered to participate in this study. Participants were physically active sports science students with at least two years of recreational RT experience. They were instructed to avoid any strenuous exercise over the course of the study and were also required to complete the specific RT programs without missing any session. None of them reported physical limitations, health problems or musculoskeletal injuries that could compromise tested performance. All participants were also informed of the procedures to be utilized and signed a written informed consent form prior to initiating the study. The study protocol adhered to the tenets of the Declaration of Helsinki and was approved by the Institutional Review Board.

Experimental design

A longitudinal pre-post design was employed to compare the effects of two power-oriented RT programs with different velocity loss threshold on the mechanical capacities of the

lower-body muscles. Participants were matched according to their theoretical maximum power (Samozino, Rejc, Di Prampero, Belli, & Morin, 2012) and then randomly assigned to one of the two groups: VL10 ($n = 10$, age 22.2 ± 2.5 years, body mass 73.9 ± 17.1 kg, height 174.6 ± 5.9 cm) or VL20 ($n = 10$, age 21.9 ± 1.7 years, body mass 80.1 ± 15.2 kg, height 176.7 ± 6.0 cm). Prior to the power-oriented RT program, all participants took part of six preliminary sessions (twice a week) with the purpose of ensuring a proper technique and increasing strength levels. Thereafter, they undertook two different RT programs over a four-week period (eight sessions). Both training groups performed the same exercise (loaded CMJ) at the same relative intensity (mean propulsive velocity [MPV] = $1.20 \text{ m}\cdot\text{s}^{-1}$) and training volume (36 repetitions). The only difference between the groups was the velocity loss allowed during the training sets (VL10 = 10% vs. VL20 = 20%). Participants were assessed before and after the RT program through a 15 m linear sprint and an incremental loading test in the CMJ exercise. The incremental loading test was performed 5 min after completing the linear sprint test. All sessions were conducted in the laboratory with at least 48–72 h of rest, at a consistent time of the day for each participant (± 1 h) and under similar environmental conditions ($\sim 22^\circ\text{C}$ and $\sim 60\%$ humidity).

Testing procedures

Sprint test

Three maximal 15 m indoor sprints were performed with 3 min of rest between them. The starting position was standardized with the lead-off foot placed 1 m behind the first time gate (Microgate, Bolzano, Italy). Participants were instructed to run as fast as possible and only the shortest time was selected for the subsequent analysis. The standardized warm-up consisted of jogging at a self-selected easy pace, dynamic stretching and lower-body joint mobilization exercises, followed by two sets of 15 m running progressive accelerations. The coefficient of variation (CV) and intraclass correlation coefficient (ICC) for sprint times were 2.22% (95% confidence interval [CI]: 1.68%–3.24%) and 0.71 (95% CI: 0.40–0.88).

CMJ incremental loading test

The CMJ technique involved participants standing with the knees and hips fully extended, feet approximately shoulder-width apart, and the barbell held across the top of the shoulders and upper back. Thereafter, participants initiated a downward movement until reaching 90° knee flexion, followed immediately by a jump for maximum height. To ensure the 90° knee angle, participants descended until touching an adjustable rod on a tripod with their glutei (Morales-Artacho, Padial, García-Ramos, Pérez-Castilla, & Feriche, 2017). The 90° knee angle was individually measured with a manual goniometer and the height of the tripod was maintained for all testing and training sessions. Participants were instructed to apply constant downward pressure on the barbell and to touch down the ground with extended leg in foot plantar flexion (Jiménez-Reyes et al., 2017). If any of these criteria were not met, the trial was repeated.

Table 1. Descriptive characteristics of the resistance training program.

SESSION TYPE 1					
Exercise-type	Jumping Jack	Vertical drop jump	Horizontal drop jump	Countermovement jump	Abdominal Crunch
Sets × repetitions (rest)	3 × 10 (30 s)	2 × 10 (4 min)	2 × 10 (4 min)	Sets* × 36 (4 min)	5 × 25 (30 s)
Exercise intensity		Height box: 50 cm	Height box: 50 cm	MPV: 1.20 m·s ⁻¹	
SESSION TYPE 2					
Exercise-type	Rope jump	Front box jump	Countermovement jump	Romanian deadlift	
Sets × repetitions (rest)	3 × 50 (60 s)	4 × 6 (4 min)	Sets* × 36 (4 min)	3 × 10 (2 min)	
Exercise intensity		Height box: 50 cm	MPV: 1.20 m·s ⁻¹	50% body mass	

The four-week power-oriented resistance training were the same training for both groups following the same session order (i.e. session type 1 and 2, respectively). *, the number of sets was extended until the participants performed the 36 repetitions. While the fastest MPV of the set was fixed to 1.20 m·s⁻¹ for both training groups, the VL10 stopped the set when the MPV was below 1.08 m·s⁻¹ (10% velocity loss) and the VL20 when the MPV was below 0.96 m·s⁻¹ (20% velocity loss). MPV, mean propulsive velocity.

Since all participants were able to jump correctly when loaded with 80 kg or more, the applied loads were 0.5, 20, 40, 60, and 80 kg. A 0.5 kg free bar was used in body weight condition to keep the same execution technique for all loads tested (Giroux, Rabita, Chollet, & Guilhem, 2015). Two trials per load were executed with 1 min of recovery between trials of the same load and 3 min between loads. Immediately after each jump, participants received jump height feedback to encourage them jumping as high as possible (Optojump, Microgate, Italy). Two trained spotters were present and lifting belts were used to ensure safety. The standardized warm-up included one set of five unloaded CMJ and another of three loaded CMJ with a 20 kg load. The CV and ICC for jump height were 6.35% (95% CI: 5.57–7.38) and 0.98 (95% CI: 0.98–0.99), respectively.

RT program

The characteristics of the four-week RT program are presented in Table 1. The number of training sessions (eight sessions, twice a week on non-consecutive days), exercise-type, relative intensity and volume were kept identical for both groups. The only difference between groups was the velocity loss threshold allowed during each CMJ exercise set: VL10 = 10% and VL20 = 20%.

Participants performed a total of 36 repetitions in the loaded CMJ exercise at maximal intended velocity. They received real-time velocity feedback from a linear velocity transducer (T-Force System; Ergotech, Murcia, Spain) and all sessions were supervised by researchers who verbally encouraged them to give maximal effort in each jump. Note that researchers sought for participants to complete the 36 repetitions in as few sets as possible while seeking to maximise the MPV of each repetition. When the corresponding velocity loss threshold was exceeded this was deemed a set completion and 4 min recovery period was provided before commencing the next set in order to complete the remaining repetitions to the prescribed total of 36.

The absolute load (kg) for each participant was weekly adjusted to match the intended MPV of 1.20 m·s⁻¹ (~40% 1RM). On this purpose, following the warm-up of the first training session of each week, the individual load-velocity relationship was modelled through the data provided by three external loads (20, 40, and 60 kg). Two trials were

performed at each load and inter-set rest periods were set to 3 min.

Three unloaded CMJs, separated by 30 s, were performed at the beginning (after warm-up) and the end (before cool-down) of each training session. The CMJ height loss was used to quantify the extent of fatigue induced by the training session (Jiménez-Reyes et al., 2016; Sánchez-Medina & González-Badillo, 2011). Only the highest jump height obtained pre- and post-session were kept for further analyses.

Measurement equipment and data analysis

Height (Seca 202, Seca Ltd., Hamburg, Germany) and body mass (Tanita BC-418 MA, Tanita corporation, Tokyo, Japan) were measured before and after the RT program. The loaded CMJ exercise was performed in a Smith machine (Multipower Fitness Line, Peroga, Murcia, Spain) coupled with a linear velocity transducer (T-Force System; Ergotech, Murcia, Spain) which directly sampled velocity data at a frequency of 1,000 Hz. A telemetric photocell (Microgate, Bolzano, Italy) was used for the sprint times, while an infrared platform (Optojump, Microgate, Italy) was used for the jump height measurements.

During each jump tested, mean force and velocity values were calculated from the equations proposed by Samozino, Morin, Hintzy, and Belli (2008), and validated by Jiménez-Reyes et al. (2017) for the CMJ exercise. Note that this simple field computation method has shown a high reliability (CV < 1.0% and ICC > 0.98) and excellent agreement with the same values of force ($r = 0.995$; CV = 0.7%; ICC = 1.00) and velocity ($r = 0.996$; CV = 1.4%; ICC = 1.00), as well as the force-velocity relationship parameters (range $r = 0.985$ –0.991; range CV = 1.2–7.6%; range ICC = 0.98–1.00) obtained by a force platform (Jiménez-Reyes et al., 2017). Specifically, Samozino's method only requires three simple input variables: system mass (sum of body mass + external load, in kg), jump height (in m) and push-off distance (in m). The push-off distance was a priori determined as the difference between the extended lower limb length, measured from the great trochanter to toes with maximal foot plantar flexion, and the vertical distance between the great trochanter and ground in the standardized starting position (i.e. 90° knee flexion) (Jiménez-Reyes et al., 2017). The individual value of push-off distance for each participant was maintained in both testing sessions. Thereafter, force and velocity data were modelled by a linear regression

to determine the force-velocity relationship parameters: $F[V] = F_0 - aV$, where F_0 represents the maximum force (i.e., force-intercept), V_0 is the maximum velocity (i.e., velocity-intercept), and a is the relationship slope (F_0/V_0) (Jaric, 2015). As a consequence of the force-velocity relationship being highly linear, the maximum power (P_0) was calculated as $P_0 = F_0 \cdot V_0 / 4$. Since an increase in the velocity achieved against the same absolute load is an indicator of strength improvement (Pareja-Blanco, Rodríguez-Rosell, Sánchez-Medina, Gorostiaga, & González-Badillo, 2014), the MPV achieved against each load tested (MPV₂₀, MPV₄₀, MPV₆₀, and MPV₈₀) were also analyzed. The one-repetition maximum (1RM_{est}) was estimated from the MPV attained against the 80 kg load by the following equation (Pérez-Castilla, García-Ramos, Padial, Morales-Artacho, & Ferlic, 2017): $\%1RM = -69.065 \times (MPV) + 121.342$. Only the repetition with the highest jump height or MPV at each load was considered for the force-velocity and load-velocity analyses, respectively. Finally, the jump height achieved against a free 0.5 kg barbell (JH_{0.5}) and the time to 15 m in a linear sprint (T₁₅) were analysed as athletic performance variables.

Statistical analyses

Data are presented as means and standard deviations (SD). Intra-session reliability was assessed through the CV and the ICC with the corresponding 95% CI. All reliability assessments were performed by means of a custom spreadsheet (Hopkins, 2000). Independent samples Student's t-tests were employed to assess between-group differences at baseline as well as the CMJ height loss (i.e. CMJ height post-session – CMJ height pre-session) for each training session. A two-way repeated measures ANOVA (group [VL10 vs. VL20] $\times$ time [pre-test vs. post-test]) with Bonferroni post-hoc test was conducted on each dependent variable. Statistical significance was accepted at $P < 0.05$ level. The statistical analyses were performed using the software package SPSS (IBM SPSS version 21.0, Chicago, IL, USA). Data were also assessed for clinical significance using an approach based on the magnitudes of change (Hopkins, Marshall, Batterham, & Hanin, 2009). The Cohen's d effect size (ES) with the corresponding 90% CI was calculated. Probabilities were also determined to establish whether true differences were lower, similar or higher than the smallest worthwhile change ($0.2 \times$ between-subject SD) (Boriani, Biffi, & Martignani, 2007). Quantitative chances of better or worse effects were assessed qualitatively as follows: $<1\%$, almost certainly not; 1–5%, very unlikely; 5–25%, unlikely; 25–75%, possible; 75–95%, likely; 95–99%, very likely; and $>99\%$, almost certain. If better or worse changes were both $>5\%$, the true difference was assessed as unclear (Hopkins et al., 2009). Magnitude-based inferential statistics were calculated using a custom spreadsheet (Hopkins, 2006).

Results

Comparison of the training sessions

The VL10 group performed more sets (2.7 ± 0.9 vs. 2.0 ± 0.8 ; $P < 0.001$) and less number of repetitions within each set

(13.3 ± 7.1 vs. 18.1 ± 10.2 ; $P < 0.001$) than the VL20 group. A very large between-participants variability was observed in the number of repetitions performed in the first set of the training session for both VL10 (20.0 ± 6.1 repetitions; CV = 30.3%) and VL20 groups (26.6 ± 7.8 repetitions; CV = 29.4%) (Figure 1). Participants in the VL10 group performed more repetitions at high velocities ranges (i.e., $MPV > 1.10 \text{ m} \cdot \text{s}^{-1}$: VL10 = 114.9 ± 31.5 vs. VL20 = 54.9 ± 21.4 ; $P < 0.001$), and consequently the VL20 group completed more repetitions at slow velocities ranges (i.e., $MPV < 1.10 \text{ m} \cdot \text{s}^{-1}$: VL10 = 12.2 ± 16.9 vs. VL20 = 54.6 ± 25.5 ; $P < 0.001$). Finally, the VL10 group generally showed a lower reduction in CMJ height after training than the VL20 group (VL10 = $-0.5 \pm 1.6 \text{ cm}$ vs. VL20 = $-1.8 \pm 2.6 \text{ cm}$; $P < 0.001$) (Figure 2).

Pre-post comparisons

Null-hypothesis significance testing is presented in Table 2, while intra- and inter-group magnitude-based inference statistical analyses are showed in Figure 3 and Figure 4, respectively.

Force-velocity relationship

No inter-group differences at pre-test and no significant “group $\times$ time” interactions were observed for any of the force-velocity relationship parameters. Note that after training, both groups increased P_0 due to an increase in V_0 , while F_0 remained practically unchanged.

Load-velocity relationship

No inter-group differences were observed in any of the load-velocity relationship variables at pre-test, except in MPV₂₀ that was higher for the VL20 group than VL10 group ($P = 0.043$). No significant “group $\times$ time” interactions were reported for any of the variables analyzed. However, the VL20 group showed possibly better effects with the heavy loads (MPV₆₀ and MPV₈₀) compared to VL10 group, while unclear effects were observed for the light loads (MPV₂₀ and MPV₄₀).

Athletic performance

No inter-group differences at pre-test and no significant “group $\times$ time” interactions were observed for JH_{0.5} and T₁₅ variables. While both groups significantly improved JH_{0.5}, no significant changes were observed for T₁₅.

Discussion

The present study compared the influence of two velocity loss thresholds (10% vs. 20%) during a power-oriented RT program on performance variables and mechanical capacities of lower-body muscles. Our hypothesis was rejected since the training group with a 10% velocity loss resulted in similar training response than the training group with a 20% velocity loss. As expected, the VL10 group performed significantly more repetitions at high velocities and showed smaller reductions in CMJ height after training than the VL20 group. It should be noted that a very large between-participants variability was observed in the number of repetitions completed in the first training set of the session for both groups (CV = 29.4–30.3%). As a result, this study suggests that the VBT does not seem to

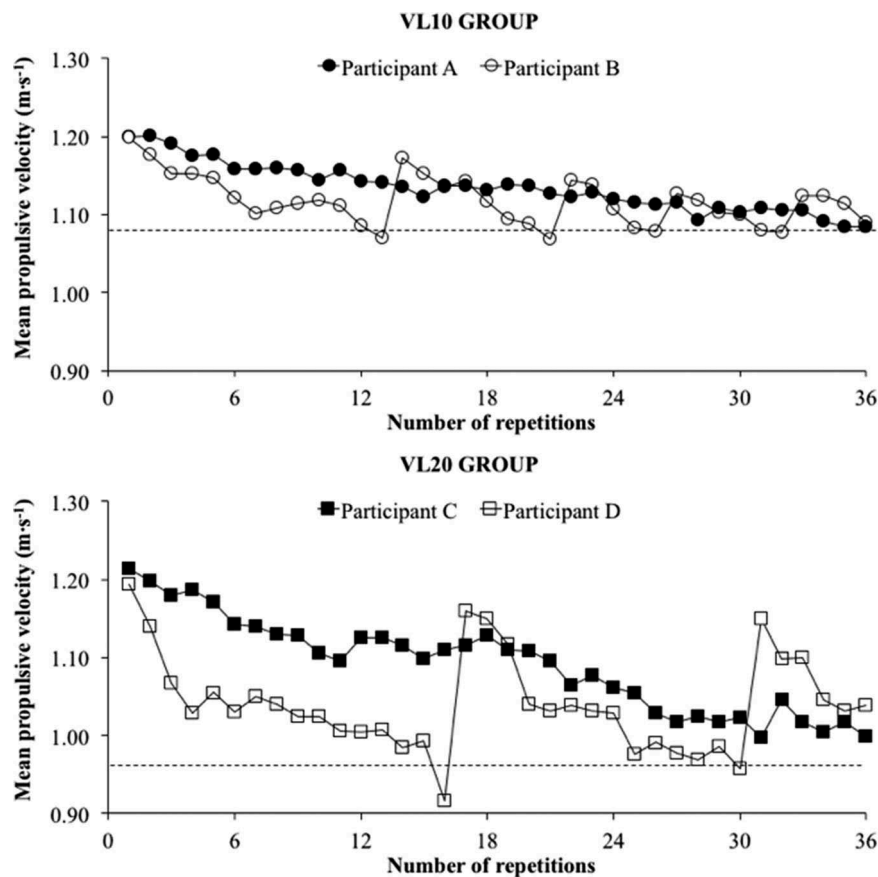


Figure 1. Velocity measurements during a training session consisting of 36 repetitions in the CMJ exercise for two participants of the VL10 group (top panel) and two participants of the VL20 group (low panel). Participant A (filled circles) completed 36 repetitions in one set with a velocity loss of 10%, participant B (open circles) completed 36 repetitions in five sets with a velocity loss of 10%, participant C (filled squares) completed 36 repetitions in one set with a velocity loss of 20%, and participant D (open squares) completed 36 repetitions in three sets with a velocity loss of 20%.

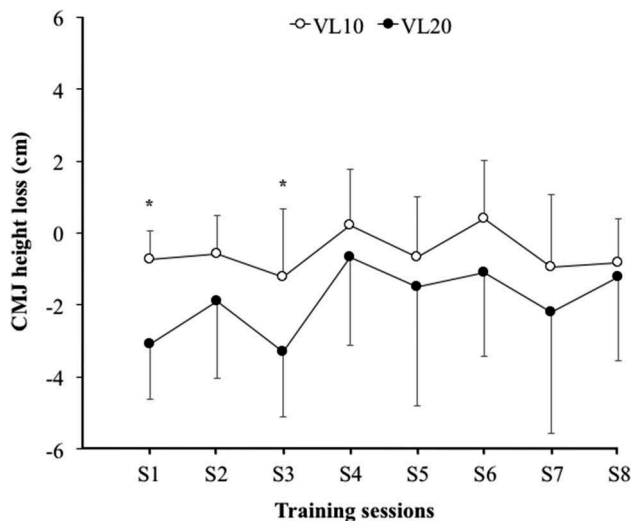


Figure 2. Comparison of CMJ height loss after each training session between the VL10 (open circles) and VL20 (filled circles) groups. Absolute differences (CMJ height post-session – CMJ height pre-session) and standard deviations are presented for the eight training sessions. *, inter-group significant differences (independent samples Student's t-tests; $P < 0.05$).

be an appropriate approach for prescribing and monitoring training volume during a power-oriented RT program conducted with the loaded CMJ exercise.

Recent studies have shown a strong relationship between the magnitude of velocity loss and the percentage of performed repetitions in the bench press and pull-up exercises ($R^2 > 0.96$ and 0.88 , respectively) (González-Badillo, Yanez-García, Mora-Custodio, & Rodríguez-Rosell, 2017; Sánchez-Moreno et al., 2017). González-Badillo et al. (2017) observed a lower between-participants variability for the percentage of repetitions completed within a given magnitude of velocity loss in each training set ($CV = 2.7\text{--}11.6\%$) compared to the exact number of repetitions actually performed against each relative load ($CV = 17.3\text{--}23.5\%$). This result seems to support the use of the VBT approach for monitoring the training volume and the actual level of effort incurred during RT programs. However, in the present study we obtained a very large between-participants variability in the number of repetitions that could be performed before reaching the 10% and 20% velocity loss thresholds ($CV = 30.3\%$ and 29.4% , respectively) in the loaded CMJ exercise. As shown in Figure 1, the number of repetitions per set performed by individuals within the same training group was highly variable. In the same figure, we can appreciate how participants A and C were able to complete the total volume of the session (36 repetitions) in one set, while participants B and D needed three and five sets, respectively. It is possible to speculate that the large variability observed in the number of repetitions completed for

Table 2. Differences in performance variables from pre- to post- resistance training program for each group.

Variables	VL10			VL20			ANOVA	
	Pre-test (mean $\pm$ SD)	Post-test (mean $\pm$ SD)	P-value	Pre-test (mean $\pm$ SD)	Post-test (mean $\pm$ SD)	P-value	Time (P-value)	Interaction (P-value)
Force-velocity relationship	F_0 (N $\cdot$ kg $^{-1}$)	33.1 $\pm$ 5.51	32.5 $\pm$ 5.68	0.252	31.5 $\pm$ 4.23	31.3 $\pm$ 3.18	0.808	0.494
	V_0 (m $\cdot$ s $^{-1}$)	3.27 $\pm$ 0.44	3.76 $\pm$ 0.61	0.067	3.46 $\pm$ 0.49	3.74 $\pm$ 0.56	0.207	0.469
	P_0 (W $\cdot$ kg $^{-1}$)	27.0 $\pm$ 4.86	30.0 $\pm$ 6.41	0.068	26.9 $\pm$ 3.35	29.2 $\pm$ 4.43	0.059	0.587
	a (N $\cdot$ s $\cdot$ kg $^{-1}$ $\cdot$ m $^{-1}$)	-10.3 $\pm$ 2.31	-8.79 $\pm$ 2.35	0.110	-9.37 $\pm$ 2.31	-8.54 $\pm$ 1.58	0.275	0.490
Load-velocity relationship	1RM $_{est}$ (kg)	148 $\pm$ 27.8	151 $\pm$ 28.1	0.322	169 $\pm$ 31.3	175 $\pm$ 27.4	0.239	0.593
	MPV $_{20}$ (m $\cdot$ s $^{-1}$)	1.54 $\pm$ 0.10	1.59 $\pm$ 0.09	0.030	1.64 $\pm$ 0.10	1.67 $\pm$ 0.12	0.227	0.613
	MPV $_{40}$ (m $\cdot$ s $^{-1}$)	1.32 $\pm$ 0.12	1.36 $\pm$ 0.12	0.053	1.40 $\pm$ 0.13	1.44 $\pm$ 0.12	0.155	0.946
	MPV $_{60}$ (m $\cdot$ s $^{-1}$)	1.13 $\pm$ 0.12	1.15 $\pm$ 0.13	0.189	1.20 $\pm$ 0.15	1.24 $\pm$ 0.13	0.063	0.215
	MPV $_{80}$ (m $\cdot$ s $^{-1}$)	0.97 $\pm$ 0.13	0.98 $\pm$ 0.13	0.587	1.03 $\pm$ 0.14	1.07 $\pm$ 0.13	0.092	0.121
Athletic performance	JH $_{0.5}$ (cm)	33.2 $\pm$ 3.66	35.3 $\pm$ 5.16	0.023	36.0 $\pm$ 5.38	37.3 $\pm$ 5.07	0.019	0.499
	T $_{15}$ (m $\cdot$ s $^{-1}$)	2.45 $\pm$ 0.11	2.46 $\pm$ 0.10	0.919	2.46 $\pm$ 0.08	2.47 $\pm$ 0.09	0.663	0.683

VL10, training group with a velocity loss threshold of 10%; VL20, training group with a velocity loss threshold of 20%; F_0 , maximum force; V_0 , maximum velocity; P_0 , maximum power; a , relationship slope; 1RM $_{est}$, estimated one-repetition maximum; MPV $_{20}$, MPV $_{40}$, MPV $_{60}$, and MPV $_{80}$, mean propulsive velocity attained against 20, 40, 60 and 80 kg loads, respectively; JH $_{0.5}$, jump height attained against free bar 0.5 kg; T $_{15}$, 15 m sprint running time.

each velocity loss threshold may be affected by the training status or the specific muscle characteristics (González-Badillo et al., 2017; Legaz-Arrese, Reverter-Masia, Munguia-Izquierdo, & Ceballos-Gurrola, 2007). However, future studies are necessary to understand the specific participant characteristics responsible for the large variability in the number of repetitions completed at a given velocity loss threshold. Therefore, we could assume that participants within the same group trained with a different degree of effort. These results suggest that the VBT approach (i.e., fixing a magnitude of velocity loss in a training set) should not be recommended over the traditional methodology (i.e., defining a priori a fixed number of repetitions per set) for prescribing the training volume during power-oriented RT programs.

To our knowledge, this is the first study analysing the effect of training with different velocity loss thresholds on the force-velocity relationship parameters (F_0 , V_0 , a , P_0). In disagreement with our hypothesis, no significant differences and *unclear* effects were observed between both training groups for all force-velocity relationship parameters. However, both training groups increased P_0 as a consequence of the increase in V_0 , while F_0 remained practically unchanged. There were also no inter-group significant differences for any of the load-velocity relationship variables (1RM $_{est}$, MPV $_{20}$, MPV $_{40}$, MPV $_{60}$, and MPV $_{80}$). Nevertheless, the practical differences analysis showed *possibly* better effects in MPV $_{60}$ and MPV $_{80}$ for VL20 group compared to VL10 group (see Figure 4). It is plausible that training with increased metabolic stress (i.e. a high velocity loss threshold; 20%) might have contributed to maximize the adaptations commonly observed following heavy loading training (Goto, Ishii, Kizuka, & Takamatsu, 2005).

Finally, our results did not show significant inter-group differences for any of the athletic performance variables (JH $_{0.5}$ and T $_{15}$). While jump height performance improved significantly after training, the 15 m sprint time did not show any change. The absence of improvements in the linear sprint is in line with previous studies that did not find sprint performance improvements in the early phase of running (i.e. acceleration phase) over short distances (< 20 m) after jump squat training with a velocity-oriented training load (< 50%1RM) (Cormie, McGuigan, & Newton, 2010; McBride, Triplett-McBride, Davie, & Newton, 2002). Coaches and strength and conditioning professionals should keep in mind that while both training groups showed similar RT responses, the VL20 group reported higher fatigue (assessed as CMJ height loss) after the training sessions, and the greater fatigue may lead to undesired training effects in the long term (Sánchez-Medina & González-Badillo, 2011).

Several limitations and directions for future research need to be addressed. First, it should be noted that the inter-repetition rest was not strictly standardized in the present study. Although participants were instructed to adopt the initial position as quickly as possible to minimize the time elapsed between successive jumps, it is possible that someone who rested longer between repetitions could

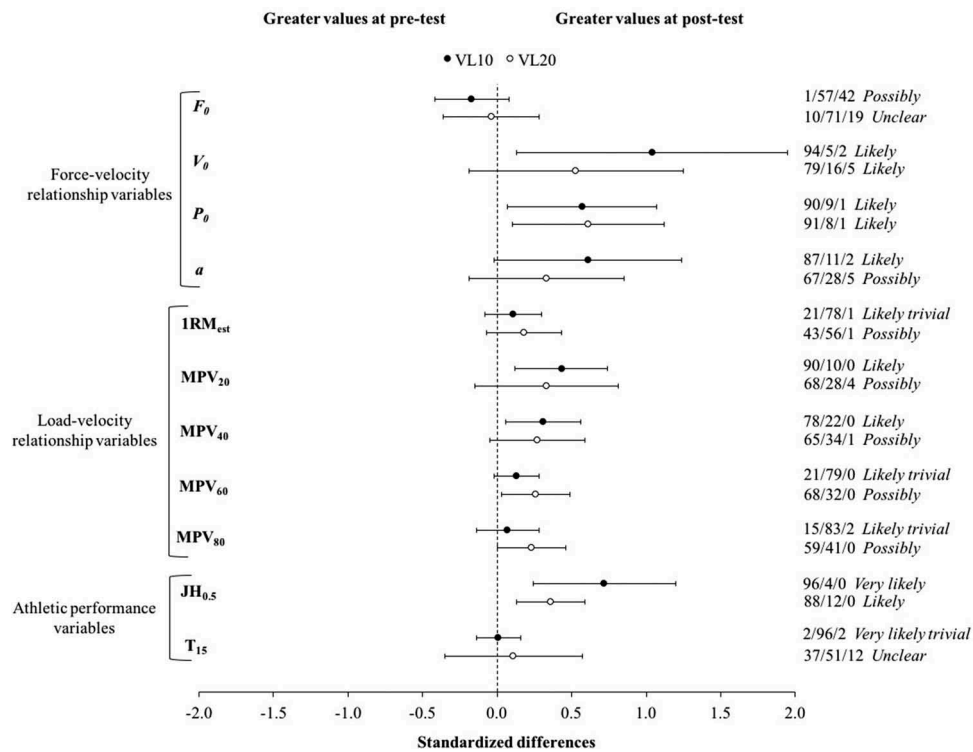


Figure 3. Standardized differences (90% confidence intervals) in all measured variables between pre- and post- test for VL10 (filled circles) and VL20 (open circles) groups. F_0 , maximum force; V_0 , maximum velocity; P_0 , maximum power; a , relationship slope; $1RM_{est}$, estimated one-repetition maximum; MPV_{20} , MPV_{40} , MPV_{60} , and MPV_{80} , mean propulsive velocity attained against 20, 40, 60 and 80 kg loads, respectively; $JH_{0.5}$, jump height attained against free bar 0.5 kg; T_{15} , 15 m sprint running time.

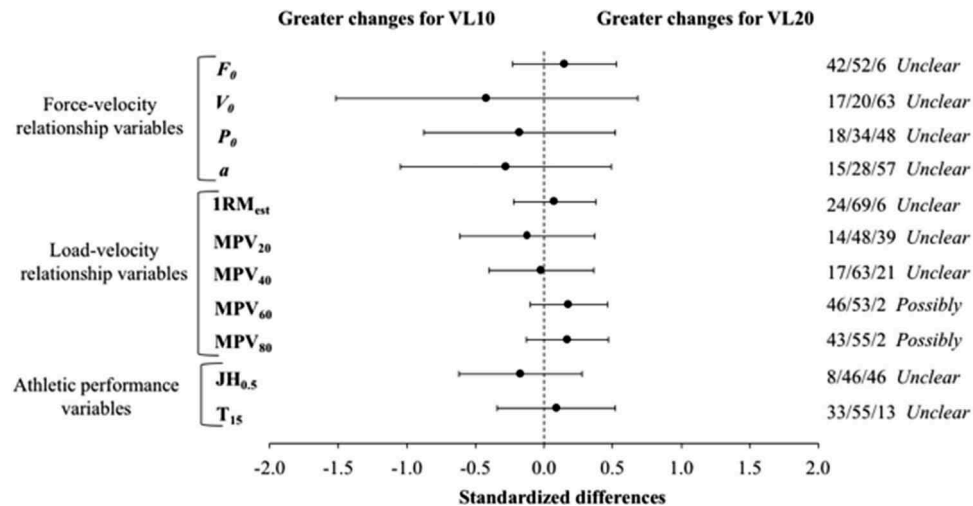


Figure 4. Standardized differences (90% confidence intervals) in all measured variables between VL10 and VL20 groups. F_0 , maximum force; V_0 , maximum velocity; P_0 , maximum power; a , relationship slope; $1RM_{est}$, estimated one-repetition maximum; MPV_{20} , MPV_{40} , MPV_{60} , and MPV_{80} , mean propulsive velocity attained against 20, 40, 60 and 80 kg loads, respectively; $JH_{0.5}$, jump height attained against an unloaded 0.5 kg barbell; T_{15} , 15 m sprint running time.

have performed more repetitions before reaching their velocity loss threshold. Thereby, future studies should be performed keeping in mind that the magnitude of velocity loss could be largely influenced by the inter-repetition rest periods. Nevertheless, the large between-participants variability in the number of repetitions completed for a given magnitude of velocity loss suggests that a combination of the

traditional (i.e., fixed number of repetitions per set) and VBT approaches could be recommended for a more precise prescription of the level of effort. In this regard, coaches can define a priori the maximum and minimum number of repetition per set, and then use the recorded velocity to decide when the set should be stopped within the desired range of repetitions.

Conclusions

The main findings of the present study revealed: 1) similar improvements for all measured variables regardless of the magnitude of the velocity loss experienced during the RT program; 2) the group that trained with a 10% velocity loss performed significantly more repetitions at high velocities and showed less degree of fatigue (i.e. CMJ height loss) than the group that trained with a 20% velocity loss; and 3) a very large between-participants variability in the number of repetitions that can be completed within a given magnitude of velocity loss. Collectively, these results question the feasibility of the VBT approach for prescribing and monitoring the repetition volume during sets of the loaded CMJ exercise, which is an exercise commonly implemented in power-oriented RT programs. It should be also noted that a 10% velocity loss largely exceeds the number of repetitions per set commonly recommended to enhance muscular power.

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Disclosure statement

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